

TEJASHVI RAJ

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EDUCATION

Vellore Institute of Technology

2022 – 2026

Bachelor of Technology - Electronics and Communication Engineering; CGPA: 8.63/10 Bhopal, Madhya Pradesh

SKILLS

Programming : C++, Embedded C, Basic Python

Tools and Platforms : ESP32, Arduino, Raspberry Pi, KiCad, Blynk IoT, LTSpice, Tinkercad, SIMULINK, GitHub

Core Areas : Embedded Systems, PCB Design, Electronics Hardware, Industrial Automation Fundamentals, VLSI Design

EXPERIENCE

Sensors and Controls Intern

Jul. 2026 – Present

Seisou Labs Private Limited

Bengaluru, Karnataka

- Designed a custom ESP32-S3 embedded PCB in KiCad, integrating multi-sensor interfaces, power management, and generating manufacturing-ready Gerber files.
- Evaluated embedded controller architectures, communication methods, and hardware reliability for an industrial radiant cooling system.
- Performed sensor integration, prototype validation, and technical documentation to support reliable and energy-efficient product development.

Project Intern

Nov. 2025 – Dec. 2025

Shitashii Innovations Private Limited

IIT Kanpur, UP

- Contributed to the development of a wearable ECG and EMG monitoring device using embedded hardware, analog signal conditioning, ESP32, and Raspberry Pi.
- Assisted in embedded hardware integration, Raspberry Pi interfacing, biomedical signal acquisition, and system debugging.
- Designed PCB schematics and layouts using KiCad while supporting hardware validation, component-level debugging, and circuit optimization.

PROJECTS

DHARA : The Ultimate Aqua-Hydro Manager | [View Project](#)

Embedded Systems, IoT

- Developed an ESP32-based autonomous monitoring and control system for hydroponic and aquaponic environments integrating TDS, pH, turbidity, and temperature sensing with automated pH regulation and cloud analytics.
- Implemented embedded firmware, multi-sensor integration, sensor calibration, and UART communication between ESP32 and Arduino Uno using C++ and Ubidots IoT for real-time monitoring.

Smart Bin: IoT-Based Touchless Dustbin | [View Project](#)

Embedded Systems, IoT

- Designed an ESP32-based touchless smart waste management system using ultrasonic sensing and servo motor control for automatic lid operation, real-time overflow detection, and cloud notifications.
- Integrated sensors, optimized embedded firmware, and implemented IoT connectivity using ESP32, Blynk IoT, and C++ to improve system responsiveness and reliability.

PCB Design Lab | [View Repository](#)

PCB Design / Embedded Hardware Development

- Designed and developed embedded hardware and custom PCB projects using KiCad, including schematic capture, PCB layout, routing, DRC verification, and Gerber generation.
- Developed manufacturing-ready PCB designs through component selection, footprint creation, placement optimization, routing, and design verification following industry-standard PCB design practices.

CO-CURRICULAR

Co-Head, Electronics Team, VITroniX Club

Jul. 2024 – Feb. 2025